FHRP: First Hop Redundancy Protocols

A Hands-On Guide to Redundant Routing

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0.1 Introduction to Network Topology for FHRP Implementation

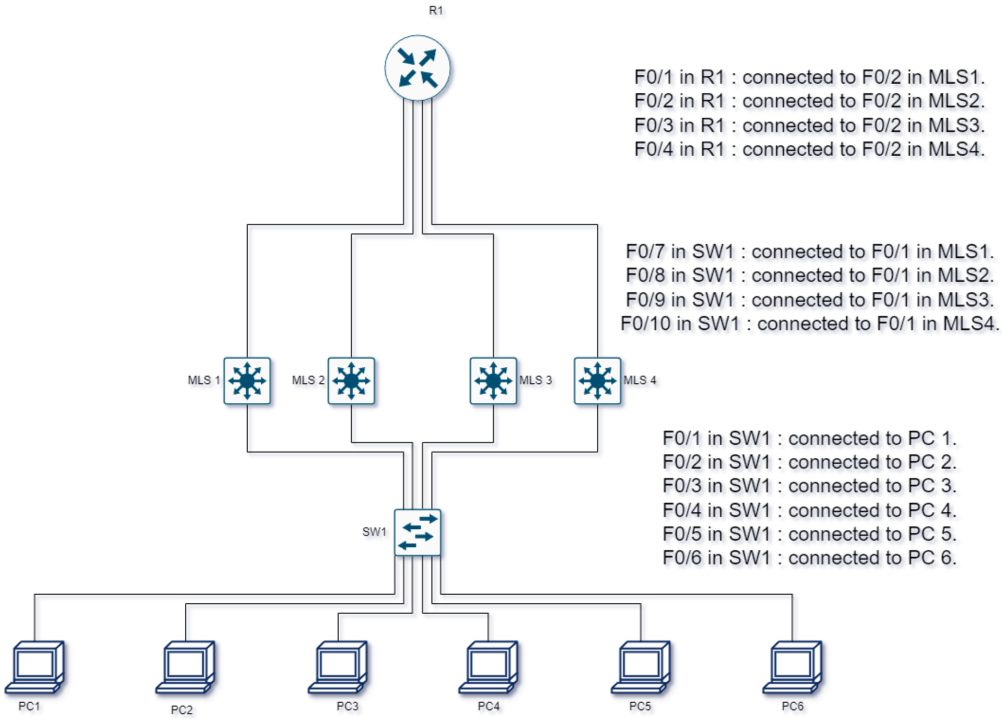

Figure 1:

This network topology is designed to demonstrate the implementation and functionality of First Hop Redundancy Protocols (FHRP), including **HSRP (Hot Standby Router Protocol)**, **VRRP (Virtual Router Redundancy Protocol)**, and **GLBP (Gateway Load Balancing Protocol)**. These protocols ensure high availability and fault tolerance by providing redundancy for default gateway routers in a network.

Topology Components

The topology consists of the following key components:

- **R1 (Router 1):** Serves as the upstream router that connects the network to external destinations.
- **MLS (Multi-Layer Switches) 1, 2, 3, and 4:** Act as Layer 3 distribution switches that participate in FHRP to provide redundancy and load balancing for end devices.
- **SW1 (Layer 2 Switch):** Functions as the access layer switch, connecting end devices (PC1 to PC6) to the network.
- **End Devices (PC1 - PC6):** Represent client machines that rely on a stable and resilient default gateway for communication beyond their local subnet.

Interconnections and Purpose

- The **end devices are connected to SW1** through FastEthernet ports (F0/1 to F0/6).
- SW1 is connected to the **multi-layer switches (MLS1 - MLS4)** via uplink ports (F0/7 to F0/10).
- Each **multi-layer switch is connected to the router (R1)** via FastEthernet links to provide redundancy at the Layer 3 distribution level.

Chapter 1

HSRP Hands-On Configuration

1.1 Introduction to HSRP Configuration

1.1.1 Overview of HSRP and Its Importance

Hot Standby Router Protocol (HSRP) is a Cisco proprietary redundancy protocol that provides **high availability** for IP networks by ensuring a backup router takes over automatically when the primary router fails.

In the **topology provided**, we have:

- R1 (Router)
- Four Multi-Layer Switches (MLS1, MLS2, MLS3, MLS4)
- One Layer 2 Switch (SW1)
- Six PCs (PC1 - PC6)

HSRP ensures **network uptime** by providing a **virtual gateway IP address**, which devices use as their default gateway. The **primary device** (Active Router) forwards packets, while the **secondary device** (Standby Router) takes over in case of failure.

1.2 Basic HSRP Setup

1.2.1 Configuring an Interface for HSRP

Each Multi-Layer Switch (MLS) needs an **HSRP virtual IP address** to act as the default gateway for connected hosts.

Lab Exercise 1: Configuring HSRP on VLAN 10

Objective: Configure **HSRP on VLAN 10** for MLS1 and MLS2 with a virtual IP.

Configuration Steps:

1. Access the VLAN interface on **MLS1** and assign an IP address.

   Click here to display the answer:

2. Repeat for **MLS2** (standby switch).

3. Click here to display the answer:

4. Verify HSRP status.

   Click here to display the answer:

Expected Outcome:

- One switch (MLS1 or MLS2) should be **Active**.
- The other switch should be **Standby**.
- The PCs should use `192.168.10.1` as the default gateway.

1.2.2 Assigning Virtual IP Addresses in HSRP

- The `standby 1 ip` command defines the **virtual IP address**.
- All hosts on **VLAN 10** should have `192.168.10.1` as their default gateway.

1.2.3 Verifying HSRP Configuration

To verify:

```
switch# show standby vlan 10
```

This command displays:

- **Active and Standby router roles**
- **Virtual IP and priority values**

1.3 HSRP Priority and Preemption

1.3.1 Setting HSRP Priority Levels

By default, the HSRP **Active Router** is chosen based on the highest **IP address**. However, we can manually assign priority.

Lab Exercise 2: Configuring HSRP Priority on MLS1 and MLS2

Objective: Set **MLS1 as the Active Router** by increasing its priority.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- MLS1 should now be the **Active Router** because it has **higher priority (110 vs. 100).**
- MLS2 should remain **Standby**.

1.3.2 Configuring Preemption for Faster Failover

- If the Active Router (**MLS1**) fails, **MLS2** takes over.
- When MLS1 comes **back online**, it should **automatically reclaim the Active role**. This is done using **preemption**.

Lab Exercise 3: Configuring Preemption

Objective: Ensure **MLS1 reclaims its role** as Active after recovery.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If MLS1 fails, MLS2 becomes Active.
- When MLS1 returns, it **automatically reclaims** the role because of preemption.

1.3.3 Adjusting Preemption Delay Timers

To avoid flapping (switching back and forth between routers), we can add a **preemption delay** to wait before re-electing the Active Router.

Lab Exercise 4: Configuring Preemption Delay

Objective: Set a **30-second delay** before preemption occurs.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- **MLS1 will wait 30 seconds** before reclaiming Active status after recovering from failure.

1.4 HSRP Timer Adjustments

1.4.1 Modifying HSRP Hello and Hold Timers

By default, HSRP uses the following timers:

- **Hello Timer**: 3 seconds
- **Hold Timer**: 10 seconds

Adjusting these timers can **speed up failover** or **stabilize the network** in case of high traffic or latency.

Lab Exercise 5: Adjusting HSRP Timers

Objective: Configure **faster failover** by modifying **hello** and **hold timers** on VLAN 10.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- **Hello Timer = 1 second**
- **Hold Timer = 3 seconds**
- **Failover will happen faster** if the Active Router fails.

1.5 HSRP Authentication

HSRP authentication prevents **unauthorized devices** from participating in the election process, which enhances security.

1.5.1 Configuring Text-Based Authentication

Text authentication allows devices in the same group to verify each other using a **shared password**.

Lab Exercise 6: Configuring Text Authentication

Objective: Secure HSRP using a **shared password ("SecureHSRP")**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- Both switches will **only communicate** if they have the same authentication key.

1.5.2 Configuring MD5 Authentication

MD5 authentication encrypts the authentication key, making it more secure than plain text.

Lab Exercise 7: Configuring MD5 Authentication

Objective: Use **MD5 authentication** to secure HSRP communications.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- HSRP **will not function** if authentication keys don't match.

1.5.3 Configuring MD5 Key-Chain Authentication

Instead of storing a key as a plain string, we can use **Key-Chains**, which allow key rotation for added security.

Lab Exercise 8: Configuring Key-Chain Authentication

Objective: Use **MD5 key-chain authentication** for HSRP.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- HSRP **uses encrypted authentication** with rotating keys.

1.6 HSRP Tracking Mechanisms

HSRP tracking allows switches to **monitor interface states** and adjust priorities dynamically when an interface fails.

1.6.1 Tracking Interface Status with HSRP

When an uplink fails, the **standby router should take over immediately**.

Lab Exercise 9: Configuring HSRP Interface Tracking

Objective: Track **interface GigabitEthernet 1/1** on MLS1. If it fails, reduce priority by **20** to allow MLS2 to take over.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If `GigabitEthernet1/1` goes down, MLS1's priority decreases by **20**.
- If priority drops below MLS2, MLS2 will become **Active Router**.

1.6.2 Using Object Tracking for Failover

Instead of tracking interfaces, we can track an **IP route or other conditions**.

Lab Exercise 10: Configuring Object Tracking for HSRP

Objective: Track an **IP route (192.168.1.0/24)** and decrement HSRP priority if it becomes unreachable.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If `192.168.1.0/24` route fails, MLS1 will lower its priority by **15**, allowing MLS2 to take over.

1.6.3 Adjusting HSRP Tracking Decrement Values

You can modify **how much priority is reduced** when tracking a failure. For example, a higher decrement value makes failover happen **faster**:

```
MLS1(config-if)# standby 1 track 1 decrement 30
```

Expected Outcome:

- If the tracked object **fails**, the priority is lowered more significantly, triggering a failover faster.

1.7 HSRP Versioning and IPv6

1.7.1 Upgrading to HSRP Version 2

By default, HSRP uses **Version 1**, but **Version 2** offers:

- Support for **larger group numbers** (0–4095 instead of 0–255)
- IPv6 compatibility
- Faster failover performance

Lab Exercise 11: Enabling HSRP Version 2

Objective: Upgrade HSRP to **Version 2** on MLS1 and MLS2.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- **HSRP version 2** should be active on both switches.
- **Virtual IP and group number remain unchanged.**

1.7.2 Configuring HSRP for IPv6

HSRP also works with **IPv6**, providing the same high-availability benefits.

Lab Exercise 12: Configuring HSRP for IPv6

Objective: Implement **IPv6 HSRP** on VLAN 20 using **a link-local virtual address**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- IPv6 HSRP should be operational.
- **Virtual IPv6 address** should be auto-configured.

1.8 Verifying and Troubleshooting HSRP

1.8.1 Using `show standby` Commands for Verification

The `show standby` command is essential for **monitoring HSRP status**.

Lab Exercise 13: Verifying HSRP Configuration

Objective: Check **which router is active and standby** in VLAN 10. **Verification Command:**

Click here to display the answer:

Expected Output Example:

```
Vlan10  Group 1  State: Active
Virtual IP: 192.168.10.1
Active Router: Local
Standby Router: 192.168.10.3
Priority: 110
Preemption: Enabled
```

- **Active Router:** MLS1 (local device).
- **Standby Router:** MLS2 (192.168.10.3).
- **Preemption enabled** to reclaim Active role when MLS1 recovers.

Other Useful Commands:

1. **Show detailed HSRP information:**

```
MLS1# show standby vlan 10
```

2. **Check timers and authentication settings:**

```
MLS1# show standby vlan 10 detail
```

1.8.2 Common HSRP Misconfigurations and Fixes

Issue #1: Both Routers in Active State

Problem: Both MLS1 and MLS2 are showing as **Active**. **Solution:**

- Ensure **HSRP authentication keys match**.

```
MLS1# show standby vlan 10 | include Auth
```

- If different, reconfigure authentication:

```
MLS1(config-if)# standby 1 authentication md5 key-string HSRP_MD5
MLS2(config-if)# standby 1 authentication md5 key-string HSRP_MD5
```

Issue #2: Standby Router Never Becomes Active

Problem: When MLS1 fails, **MLS2 does not take over**. **Solution:**

- Ensure **preemption is enabled**:

```
MLS2(config-if)# standby 1 preempt
```

- Check if interface tracking is **causing a low priority**.

```
MLS2# show standby vlan 10
```

  If priority is **too low**, increase it:

```
MLS2(config-if)# standby 1 priority 120
```

Issue #3: Hosts Can't Reach the Virtual Gateway

Problem: PCs cannot ping `192.168.10.1 (HSRP virtual IP)`. **Solution:**

- Check if VLAN **interface is up** on both switches:

```
MLS1# show ip interface brief | include Vlan10
```

- If it shows **"administratively down"**, enable it:

```
MLS1(config)# interface vlan 10
MLS1(config-if)# no shutdown
```

Chapter 2

VRRP Hands-On Configuration

2.1 Introduction to VRRP

2.1.1 Overview of VRRP and Its Importance

The **Virtual Router Redundancy Protocol (VRRP)** is an open-standard redundancy protocol that provides high availability for **default gateways** by electing a **Master** and **Backup** router.

VRRP ensures that if the **Master Router fails**, a **Backup Router** takes over **seamlessly**, preventing network downtime.

In the **topology provided**, we have:

- Router (R1)
- Four Multi-Layer Switches (MLS1, MLS2, MLS3, MLS4)
- One Layer 2 Switch (SW1)
- Six PCs (PC1 - PC6)

VRRP provides **fault tolerance** by allowing multiple **Layer 3 switches** to participate in the election process while hosts continue using a single **virtual IP address** as the default gateway.

2.1.2 Basic VRRP Concepts and Terminology

- **Master Router**: The **active** VRRP router that handles traffic.
- **Backup Router**: The **standby** VRRP router that takes over if the Master fails.
- **Virtual Router**: The **group of devices** sharing a single **virtual IP**.
- **Virtual IP Address (VIP)**: The **gateway address** used by end devices.
- **VRRP Group**: Identifies multiple routers participating in the **same failover group**.

2.2 Basic VRRP Setup

2.2.1 Configuring an Interface for VRRP

Each **Multi-Layer Switch (MLS)** needs to participate in a **VRRP group** with a **virtual IP address** for failover.

Lab Exercise 1: Configuring VRRP on VLAN 10

Objective: Configure **VRRP on VLAN 10** for MLS1 and MLS2 with a **virtual IP**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- One switch (MLS1 or MLS2) should become the **Master**.
- The other switch should be **Backup**.
- PCs should use **192.168.10.1** as their **default gateway**.

2.2.2 Assigning Virtual IP Addresses in VRRP

- The `vrrp 1 ip` command defines the **virtual IP address** shared by the VRRP group.
- Hosts in VLAN 10 will use this **virtual IP** (`192.168.10.1`) as their default gateway.

2.2.3 Verifying VRRP Configuration

To check VRRP status:

```
MLS1# show vrrp vlan 10
```

This command displays:

- **Master and Backup router roles**
- **Virtual IP and priority values**
- **VRRP timers**

2.3 VRRP Priority and Master Election

2.3.1 Understanding VRRP Priority Levels

VRRP uses a **priority system (1–255)** to determine the **Master Router**.

- **Default priority = 100**
- **Highest priority wins**
- **Master router** always has the highest priority

2.3.2 Configuring VRRP Priority for Master Selection

By default, the router with the **highest IP address** becomes **Master**, but you can manually set priority.

Lab Exercise 2: Configuring VRRP Priority on MLS1 and MLS2

Objective: Ensure **MLS1 becomes the Master Router** by assigning it a higher priority.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- MLS1 should now be the **Master Router** because its **priority is 110 (higher than MLS2's 100).**
- MLS2 should remain **Backup Router**.

2.3.3 VRRP Master Election Process

VRRP determines the Master using these steps:

1. **Highest priority wins.**
2. If equal, the **router with the highest IP address wins**.
3. If the Master fails, the **Backup takes over** automatically.

Verifying Master Router:

```
MLS1# show vrrp
```

Expected Output Example:

```
Vlan10  Group 1  State: Master
Virtual IP: 192.168.10.1
Priority: 110
Preemption: Enabled
Master Router: Local
```

2.4 VRRP Preemption and Timers

2.4.1 Enabling Preemption for Faster Failover

By default, **preemption is enabled** in VRRP. This means that if a **higher-priority router** comes back online, it will **automatically reclaim the Master role**.

Lab Exercise 3: Configuring VRRP Preemption

Objective: Ensure **MLS1 automatically becomes Master** when it recovers.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If **MLS1 fails**, MLS2 will take over.
- When MLS1 **comes back online**, it will reclaim the Master role automatically.

2.4.2 Adjusting VRRP Advertisement Timers

VRRP uses these timers:

- **Advertisement Interval**: How often the Master sends VRRP hello messages (default 1 second).
- **Hold Time**: How long the Backup waits before assuming Master role (default 3 seconds).

Lab Exercise 4: Configuring Custom VRRP Timers

Objective: Reduce **failover time** by setting **faster VRRP timers**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- Failover happens **within 1.5 seconds** instead of the default 3 seconds.

2.5 VRRP Authentication

2.5.1 Configuring Text-Based Authentication in VRRP

VRRP authentication ensures only **authorized routers** participate in the election.

Lab Exercise 5: Configuring VRRP Authentication

Objective: Secure VRRP using a **shared password ("SecureVRRP")**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- Only devices with the **correct authentication key** will communicate.

2.5.2 Implementing MD5 Authentication for VRRP

MD5 authentication is **more secure** because it encrypts the authentication key.

Lab Exercise 6: Configuring MD5 Authentication

Objective: Use **MD5 authentication** to secure VRRP communications.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- VRRP **will not function** if authentication keys don't match.

2.6 VRRP Object Tracking and Failover

2.6.1 Tracking Interface Status with VRRP

VRRP **can track interfaces** and **reduce priority** if a tracked interface fails. This ensures a **Backup Router** takes over immediately.

Lab Exercise 7: Configuring VRRP Interface Tracking

Objective: Track **interface GigabitEthernet 1/1** on MLS1. If it fails, reduce priority by **20**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If `GigabitEthernet1/1` goes down, MLS1's priority decreases by **20**.
- If priority drops below MLS2, MLS2 will become **Master Router**.

2.6.2 Using Object Tracking to Adjust Priority Dynamically

Instead of tracking interfaces, VRRP can track an **IP route** and adjust priority dynamically.

Lab Exercise 8: Configuring IP Route Tracking for VRRP

Objective: Track an **IP route (192.168.1.0/24)** and decrement VRRP priority if it becomes unreachable.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If the **tracked route fails**, MLS1 will lower its priority by **15**, allowing MLS2 to take over.

2.6.3 Adjusting VRRP Tracking Decrement Values

You can modify **how much priority is reduced** when tracking a failure.

For example, a **higher decrement value** speeds up failover:

```
MLS1(config-if)# vrrp 1 track 1 decrement 30
```

Expected Outcome:

- If the tracked object **fails**, the priority is lowered more significantly, triggering a failover faster.

2.7 VRRP Versioning and IPv6 Support

2.7.1 Upgrading to VRRP Version 3

VRRP **Version 3** supports **IPv6** and provides better scalability.

Key Benefits of VRRPv3:

- **Supports IPv6**
- Allows **higher group numbers** (up to 4095)
- Improves **failover performance**

Lab Exercise 9: Upgrading VRRP to Version 3

Objective: Upgrade VRRP to **Version 3** on MLS1 and MLS2.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- VRRP **Version 3** should be active on both switches.
- **Virtual IP and group number remain unchanged**.

2.7.2 Configuring VRRP for IPv6 High Availability

VRRPv3 allows redundancy for **IPv6 networks**. Instead of a **virtual IPv4 address**, it assigns a **virtual IPv6 address**.

Lab Exercise 10: Configuring VRRP for IPv6

Objective: Implement **IPv6 VRRP** on VLAN 20 using a **link-local virtual address**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- IPv6 VRRP should be operational.
- **Virtual IPv6 address (fe80::1)** should be reachable.

2.7.3 Verifying and Troubleshooting VRRPv3

To verify VRRP for IPv6:

```
MLS1# show vrrp all
```

Expected Output:

```
Vlan20  Group 1  State: Master
Virtual IPv6: fe80::1
Priority: 110
Master Router: Local
```

2.8 Verifying and Troubleshooting VRRP

2.8.1 Using `show vrrp` Commands for Verification

The `show vrrp` command is essential for **monitoring VRRP status**.

Lab Exercise 11: Verifying VRRP Configuration

Objective: Check **which router is Master and Backup** in VLAN 10.

Verification Command:

Click here to display the answer:

Expected Output Example:

```
Vlan10  Group 1  State: Master
Virtual IP: 192.168.10.1
Master Router: Local
Backup Router: 192.168.10.3
Priority: 110
Preemption: Enabled
```

- **Master Router:** MLS1 (local device).
- **Backup Router:** MLS2 (192.168.10.3).
- **Preemption enabled** to reclaim the Master role.

Other Useful Commands:

1. **Show detailed VRRP information:**

```
MLS1# show vrrp vlan 10
```

2. **Check timers and authentication settings:**

```
MLS1# show vrrp vlan 10 detail
```

2.8.2 Common VRRP Misconfigurations and Fixes

Issue #1: Both Routers in Master State

Problem: Both MLS1 and MLS2 are showing as **Master**.
Solution:

- Ensure **VRRP authentication keys match**.

```
MLS1# show vrrp vlan 10 | include Auth
```

- If different, reconfigure authentication:

```
MLS1(config-if)# vrrp 1 authentication md5 key-string VRRP_MD5
MLS2(config-if)# vrrp 1 authentication md5 key-string VRRP_MD5
```

Issue #2: Backup Router Never Becomes Master

Problem: When MLS1 fails, **MLS2 does not take over**.
Solution:

- Ensure **preemption is enabled**:

```
MLS2(config-if)# vrrp 1 preempt
```

- Check if **interface tracking is causing a low priority**.

```
MLS2# show vrrp vlan 10
```

If priority is **too low**, increase it:

```
MLS2(config-if)# vrrp 1 priority 120
```

Issue #3: Hosts Can't Reach the Virtual Gateway

Problem: PCs cannot ping `192.168.10.1 (VRRP virtual IP)`.
Solution:

- Check if VLAN **interface is up** on both switches:

```
MLS1# show ip interface brief | include Vlan10
```

- If it shows **"administratively down"**, enable it:

```
MLS1(config)# interface vlan 10
MLS1(config-if)# no shutdown
```

Chapter 3

GLBP Hands-On Configuration

3.1 Introduction to GLBP

3.1.1 Overview of GLBP and Its Benefits

Gateway Load Balancing Protocol (**GLBP**) is a **Cisco-proprietary** redundancy protocol that provides both **gateway failover** and **load balancing** for network traffic. Unlike **HSRP** and **VRRP**, which only elect one active router at a time, GLBP enables multiple routers to **simultaneously** handle network traffic while offering redundancy.

Key Benefits of GLBP:

- **Load Balancing:** Distributes network traffic across multiple routers.
- **Redundancy:** Ensures **automatic failover** if one router fails.
- **Active Virtual Gateway (AVG):** Manages **multiple Active Virtual Forwarders (AVFs)** for traffic distribution.

Network Topology Overview

In our topology, we have:

- Router R1
- Four Multi-Layer Switches (MLS1, MLS2, MLS3, MLS4)
- One Layer 2 Switch (SW1)
- Six PCs (PC1 - PC6)

Using **GLBP**, we will configure **MLS1 and MLS2** as **gateway routers**, ensuring that:

1. **Traffic is load-balanced** across multiple routers.
2. **Failover occurs automatically** if one router fails.

3.1.2 Differences Between GLBP, HSRP, and VRRP

Feature	**GLBP**	**HSRP**	**VRRP**
Load Balancing	✓ Yes	× No	× No
Master/Active Role	Multiple Active Virtual Forwarders	Single Active Router	Single Master Router
Preemption	✓ Enabled by Default	✓ Configurable	✓ Enabled by Default
IPv6 Support	✓ Yes (GLBPv6)	✓ Yes (HSRPv2)	✓ Yes (VRRPv3)
Object Tracking	✓ Yes	✓ Yes	✓ Yes

3.1.3 Basic GLBP Concepts and Terminology

- **Active Virtual Gateway (AVG)**: The router that assigns virtual MAC addresses to participating routers.
- **Active Virtual Forwarder (AVF)**: Routers that **actively forward traffic** based on assigned MAC addresses.
- **GLBP Group**: A set of routers configured together for redundancy.
- **Virtual IP Address**: The single IP address shared across multiple routers in the GLBP group.

3.2 Basic GLBP Setup

3.2.1 Configuring an Interface for GLBP

Each **Multi-Layer Switch (MLS)** needs to participate in a **GLBP group** with a **virtual IP address** for failover and load balancing.

Lab Exercise 1: Configuring Basic GLBP on VLAN 10

Objective: Configure **GLBP on VLAN 10** for **MLS1 and MLS2** using a shared virtual IP.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- One switch (MLS1 or MLS2) will become the **AVG**.
- Both switches will act as **AVFs**, handling network traffic.
- PCs should use `192.168.10.1` as their default gateway.

3.2.2 Assigning Virtual IP Addresses in GLBP

- The `glbp 1 ip` command defines the **virtual IP address** shared by the GLBP group.
- Hosts in VLAN 10 will use `192.168.10.1` as their **default gateway**.

3.2.3 Verifying GLBP Configuration

To check GLBP status:

```
MLS1# show glbp vlan 10
```

Expected Output:

```
Interface: Vlan10
Group: 1
State: Active
Virtual IP: 192.168.10.1
Active Virtual Gateway: Local
```

This command displays:

- **Active Virtual Gateway (AVG)**
- **Active Virtual Forwarders (AVFs)**
- **Load Balancing Status**

3.3 GLBP Load Balancing Methods

3.3.1 Understanding GLBP Load Balancing Mechanisms

GLBP supports three **load-balancing** methods:

Load Balancing Mode	**Description**
Round-Robin	Default mode – traffic is equally distributed across AVFs.
Weighted	Assigns traffic based on router weights (priority).
Host-Dependent	Ensures that each host always uses the same gateway.

3.3.2 Configuring Different Load Balancing Methods

Lab Exercise 2: Configuring Load Balancing on VLAN 10

Objective: Configure **different GLBP load balancing methods** on MLS1 and MLS2.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- **Traffic should be distributed based on the chosen method.**
- **GLBP will balance traffic dynamically across routers.**

3.3.3 Monitoring and Verifying Load Balancing Performance

To monitor **GLBP traffic distribution**, use:

```
MLS1# show glbp vlan 10 detail
```

To check **GLBP weighting and failover status**, use:

```
MLS1# show glbp 1
```

Expected Output Example:

```
Vlan10 - Group 1 - Active Virtual Gateway
Load Balancing: Weighted
Active Virtual Forwarders: 2
Virtual MAC: 0007.b400.0101
```

- **GLBP should distribute traffic** based on the configured method.
- **Failover should happen dynamically** if an AVF fails.

3.4 GLBP Priority and Active Virtual Gateway (AVG) Election

3.4.1 Understanding GLBP Priority and AVG Role

GLBP determines the **Active Virtual Gateway (AVG)** based on **priority values**. The AVG is responsible for assigning **Virtual MAC addresses** to the Active Virtual Forwarders (AVFs).

- The router with the **highest priority** becomes the **AVG**.
- If the AVG fails, the router with the **next highest priority** takes over.
- Default GLBP priority is **100**, but it can be **manually adjusted**.

3.4.2 Configuring GLBP Priority for AVG Selection

Lab Exercise 3: Configuring GLBP Priority on VLAN 10

Objective: Ensure **MLS1 becomes the AVG** by assigning it a higher priority.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- MLS1 should be the **AVG** because its **priority is higher** than MLS2's 100.
- MLS2 will remain as a **backup AVG**.

3.4.3 Preemption and AVG Failover Mechanism

If the AVG **fails**, another router takes over. However, when the **original AVG recovers**, we want it to reclaim its role. This is called **Preemption**.

Lab Exercise 4: Configuring GLBP Preemption

Objective: Ensure **MLS1 automatically reclaims AVG** role after failure.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If MLS1 fails, MLS2 will **take over as AVG**.
- When MLS1 **comes back online**, it **automatically reclaims AVG status**.

3.5 GLBP Authentication and Security

3.5.1 Configuring Plain Text Authentication in GLBP

Authentication in GLBP ensures that only **authorized routers** participate in GLBP elections.

Lab Exercise 5: Configuring Text Authentication on MLS1 and MLS2

Objective: Secure GLBP using a **shared password ("SecureGLBP")**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- Only routers with **matching authentication keys** will participate in GLBP elections.

3.5.2 Implementing MD5 Authentication for Secure GLBP Communications

MD5 authentication encrypts the password, making it more secure than plain text authentication.

Lab Exercise 6: Configuring MD5 Authentication on MLS1 and MLS2

Objective: Secure GLBP using **MD5 authentication**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- GLBP will **not function** if authentication keys don't match.

3.6 GLBP Tracking and Failover

3.6.1 Tracking Interfaces to Adjust Priority Dynamically

GLBP can monitor **interface health** and **reduce priority dynamically** if a tracked interface goes down.

Lab Exercise 7: Configuring GLBP Interface Tracking

Objective: Track **interface GigabitEthernet1/1** on MLS1. If it fails, reduce priority by **20**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If `GigabitEthernet1/1` fails, MLS1's priority will **decrease by 20**.
- If priority drops below MLS2's, MLS2 will become **the new AVG**.

3.6.2 Using Object Tracking to Enhance Redundancy

Instead of tracking interfaces, **GLBP can track routes** and reduce priority if a critical route disappears.

Lab Exercise 8: Configuring Route Tracking for GLBP

Objective: Track an **IP route (192.168.1.0/24)** and decrement GLBP priority if it becomes unreachable.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If the **192.168.1.0/24 route fails**, MLS1 will lower its GLBP priority by **15**, allowing MLS2 to take over.

3.6.3 Configuring GLBP Weighting for Dynamic Failover

GLBP can use **weight values** to determine when a router should stop forwarding traffic.

Lab Exercise 9: Configuring GLBP Weighting

Objective: Set up **GLBP weighting** to dynamically **disable forwarding** if a router becomes unstable.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- If MLS1's **weight drops below 90**, it **stops forwarding traffic**.
- If the weight **recovers above 120**, it **resumes forwarding traffic**.

3.7 GLBP Timers and Optimization

3.7.1 Adjusting GLBP Hello and Hold Timers

GLBP uses **timers** to control how often the Active Virtual Gateway (AVG) sends **hello messages** and how long a Backup router should wait before taking over.

Default GLBP Timers:

- **Hello Timer:** 3 seconds
- **Hold Timer:** 10 seconds

Lab Exercise 10: Configuring Custom GLBP Timers

Objective: Reduce **failover time** by modifying **hello** and **hold timers**.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- **Failover will happen within 4 seconds** instead of the default 10 seconds.

3.7.2 Configuring Redirect and Timeout Timers

GLBP includes **redirect and timeout timers** to optimize traffic switching during failover.

Key GLBP Timer Settings:

- **Redirect Timer:** How long the AVG allows traffic to flow through an AVF before redirecting it.
- **Timeout Timer:** When the MAC address of a failed AVF is removed from the GLBP group.

Lab Exercise 11: Configuring GLBP Redirect Timers

Objective: Optimize **GLBP failover performance** by adjusting redirect and timeout timers.

Configuration Steps:

Click here to display the answer:

Expected Outcome:

- **Traffic redirection** will occur within 10 minutes of an AVF failure.
- After 15 minutes, GLBP **removes the failed AVF's MAC address**.

3.7.3 Optimizing GLBP for Network Performance

To further optimize GLBP, consider:

- ✓ **Using weighted load balancing** for intelligent traffic distribution.
- ✓ **Fine-tuning timers** to meet network stability needs.
- ✓ **Configuring object tracking** to dynamically adjust priorities.

To check **GLBP performance**, use:

```
MLS1# show glbp detail
```

3.8 Verifying and Troubleshooting GLBP

3.8.1 Using `show glbp` Commands for Verification

The `show glbp` command provides critical information about **GLBP roles, timers, and load balancing status**.

Lab Exercise 12: Checking GLBP Status

Objective: Verify **GLBP operation and identify the AVG and AVFs**.

Verification Commands:

Click here to display the answer:

Expected Output Example:

```
Interface  Group  Role    State  Priority  Preempt  Virtual IP
Vlan10      1     AVG     Active  110       Yes      192.168.10.1
Vlan10      1     AVF     Standby 100       Yes      192.168.10.1
```

This command confirms:

- ✓ Which router is the **AVG**
- ✓ Which routers are **AVFs**
- ✓ Current **priority levels**

3.8.2 Debugging GLBP Load Balancing and Failover Issues

When **GLBP does not work correctly**, consider the following common issues:

Issue #1: GLBP Not Load Balancing Traffic

Problem: Traffic is only using one router.
Solution: Ensure **GLBP load balancing is enabled**:

```
MLS1(config)# interface vlan 10
MLS1(config-if)# glbp 1 load-balancing weighted
```

Issue #2: GLBP Preemption Not Working

Problem: The higher-priority router is not reclaiming AVG.
Solution:

- Verify that **preemption is enabled**:

  ```
  MLS1(config-if)# glbp 1 preempt
  ```

- Check if **tracking is reducing the priority** below the backup router's priority:

  ```
  MLS1# show glbp vlan 10
  ```

- Manually increase the priority:

  ```
  MLS1(config-if)# glbp 1 priority 120
  ```

Issue #3: Hosts Cannot Reach the Virtual Gateway

Problem: PCs cannot ping `192.168.10.1 (GLBP Virtual IP)`.
Solution:

- Check if VLAN **interface is up** on both switches:

  ```
  MLS1# show ip interface brief | include Vlan10
  ```

- If it shows **"administratively down"**, enable it:

  ```
  MLS1(config)# interface vlan 10
  MLS1(config-if)# no shutdown
  ```

3.8.3 Common GLBP Misconfigurations and Fixes

Issue	**Cause**	**Solution**
Both routers are AVG	Incorrect priorities	Set priority on one router higher
GLBP not responding	Authentication mismatch	Ensure matching passwords
Failover takes too long	Default timers	Adjust hello and hold timers
Traffic imbalance	Default round-robin mode	Use weighted load balancing